CS-540 Data Literacy & Visualizations

4-3 Submit Assignment: Comparing Visualizations for Impact

Malgorzata Debska

Southern New Hampshire University

Prof. Mahmoud Khasawneh

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Analysis of Education Funding Visualizations

Effective data visualization plays a critical role in communicating complex information in a clear and accessible way. When multiple visualizations are present in the same data set, selecting the most proper chart becomes an important analytical task. This assignment evaluates three different visualizations that display regional education funding data and decides which is most suitable for inclusion in a public policy report intended for policymakers and education advocates. By examining how each visualization stands for the data, analyzing key characteristics, comparing their effectiveness, and recommending refinements, this analysis shows how thoughtful visualization choices can enhance understanding and support informed decision-making.

Representation of the Data

The first visualization represents regional education funding using a bar chart in which each region appears along the horizontal axis, and funding amounts appear on the vertical axis. This format allows viewers to easily compare funding levels across regions and quickly find which areas receive higher or lower levels of support. The second visualization displays the same data using a line chart, placing regions along the horizontal axis and plotting funding levels on the vertical axis with points connected by a continuous line. While this format shows variation in funding, it also suggests a progression or trend that does not naturally exist because the data is categorical rather than sequential. The third visualization uses a pie chart to illustrate how total education funding is divided among regions, with each slice being a region’s proportion of overall funding. This approach emphasizes relative share rather than exact numerical differences.

Key Characteristics of Each Visualization

Each visualization contains distinct characteristics that affect how the data is perceived. The bar chart uses a vertical layout with clearly labeled axes and a descriptive title, making it easy to interpret. Its labeling supports quick identification of regions and approximate funding values. The line chart also includes axis labels and a title, but individual data points may be harder to read, and the connected line can mislead viewers into assuming continuity between categories. The pie chart presents labeled slices with percentages or region names, but when multiple regions have similar proportions, it becomes difficult to distinguish between differences or figure out exact values.

Comparison of Visualization Effectiveness

When comparing the three visualizations, the bar chart communicates the data most clearly because it aligns well with categorical data and allows for direct side-by-side comparisons. The line chart is moderately effective but less appropriate due to its implication of trends across regions. The pie chart is the least effective for this data set because it prioritizes proportions over precise comparisons, making it harder to evaluate differences between regions. In terms of clarity and data type alignment, the bar chart best supports accurate interpretation of the data's key insights.

Most Appropriate Visualization for the Audience

The bar chart is the most right visualization for policymakers and education advocates. These audiences often need to quickly understand disparities and make informed decisions based on visible differences. The bar chart presents information in a simple, intuitive format that does not require advanced data literacy. It enables viewers to focus on policy implications rather than deciphering how the visualization works, making it the strongest choice for inclusion in the report.

Recommended Refinements

Several refinements could further improve the effectiveness of the visualizations. Adding exact data labels above each bar would allow viewers to see precise funding values without estimating the axis. Using a consistent, high-contrast color scheme would improve readability and accessibility. Sorting regions by funding amount rather than alphabetically would make disparities clearer. Simplifying titles and adding brief subtitles to indicate the year and units of measurement would provide helpful context. Including a short caption that summarizes the main takeaway would further guide interpretation.

Although all three visualizations display the same regional education funding data, their effectiveness differs considerably. The bar chart offers the clearest and most correct representation for a non-technical audience and best supports meaningful comparison. Selecting this visualization and applying targeted refinements ensures that the data is communicated in a way that is accessible, credible, and impactful, ultimately strengthening the quality of the public policy report.